# Rooms With a View: Random Dorm Assignment and Cross-Class Friendship in College

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**Abstract**

Low-income students who attend selective colleges enter environments that are, on paper, far more economically integrated than the neighborhoods they grew up in. Whether this exposure translates into cross-class friendship, and whether those friendships matter for economic outcomes, is largely unknown. We study a large public flagship university that assigns first-year students to residence-hall floors by lottery. Linking housing records for 24,871 first-year students in four entering cohorts to a campus-wide social network constructed from ID-card co-swipes, career-services records, and state wage records, we estimate the causal effect of the share of high-socioeconomic-status (SES) floor-mates on the economic connectedness (EC) of 7,412 Pell-eligible students. A 10 percentage point increase in the high-SES share of a student's floor raises her first-year EC by 0.071 (10 percent of the mean), increases the probability of holding a paid internship by the end of junior year by 2.1 percentage points, and raises earnings at age 25 by $2,350 (6.1 percent). Instrumental-variables and mediation estimates indicate that roughly 70 percent of the earnings effect operates through cross-class friendship. Effects are largest for first-generation students, Black and Hispanic students, and students from low-EC home counties. Our estimates provide the first quasi-experimental evidence that the cross-sectional relationship between economic connectedness and upward mobility documented by Chetty et al. (2022a) reflects a causal effect of cross-class friendship, and suggest that SES-balanced housing assignment is a low-cost lever for colleges seeking to increase mobility.

**JEL:** I23, I24, J62, Z13. **Keywords:** social capital, economic connectedness, peer effects, higher education, intergenerational mobility.

*Note:* This document is a teaching prop. The authors, the university, the data, and all estimates are invented. Published works in the references are real and are cited as they appear in print or in their working-paper versions. We thank seminar participants at the (fictional) Westbrook Applied Micro Lunch for comments. Correspondence: rooms-with-a-view@example.edu.

# 1 Introduction

Children who grow up in communities where low-income people have more high-income friends are much more likely to rise out of poverty. Using data on 21 billion friendships among 72.2 million Facebook users, Chetty et al. (2022a) show that *economic connectedness* (EC)—the share of above-median-SES friends among below-median-SES individuals, scaled by 50 percent—is among the strongest predictors of upward income mobility identified to date: the county-level correlation between EC and upward mobility is 0.65, and the authors estimate that if children with low-SES parents grew up in counties with EC comparable to that of the average child with high-SES parents, their incomes in adulthood would increase by 20 percent on average. Other forms of social capital, such as network cohesion and civic engagement, are only weakly related to mobility.

These associations raise a first-order question: does cross-class friendship *cause* better economic outcomes, or do high-EC places simply differ in other ways that also promote mobility? The distinction matters for policy. If friendships themselves carry information, referrals, and aspirations across class lines, then institutions that shape who meets whom—schools, colleges, workplaces, housing authorities—have a direct lever for mobility. If instead EC is a marker of unobserved community characteristics, reshuffling who lives next to whom would do little.

Colleges are a natural place to study this question. Chetty et al. (2020a) document that income segregation across colleges is as large as income segregation across neighborhoods, and conclude that "efforts to increase interaction across socioeconomic groups may be just as valuable at the college level as they are at the neighborhood level." Within a college, however, students sort into majors, clubs, and friendship groups, so low-income students at an economically diverse campus need not form cross-class ties. Housing is one of the few margins over which colleges exercise nearly complete control, and first-year residence halls are where many durable friendships begin (Marmaros and Sacerdote, 2006).

This paper exploits the random assignment of first-year students to residence-hall floors at a large public flagship university ("the University"). The University's Housing Office assigns incoming students to rooms by lottery, which generates as-good-as-random variation in the share of a student's floor-mates who come from high-SES families. We link four cohorts of housing records (entering classes of 2014–2017; 24,871 students on 628 floors) to three additional sources: (i) a campus-wide social network constructed from ID-card co-swipes, which we use to measure each student's EC in the spirit of Chetty et al. (2022a); (ii) career-services records on internships; and (iii) state unemployment-insurance (UI) wage records, which allow us to observe earnings at approximately age 25. Our analysis sample consists of the 7,412 Pell-eligible (low-SES) students in the housing lottery.

We report four main findings. First, exposure produces connection. A 10 percentage point (pp) increase in the high-SES share of a low-SES student's floor raises her first-year EC by 0.071 (s.e. 0.010), about 10 percent of the sample mean. The effect persists, at roughly 40 percent of its initial size, into the third year of college. Second, exposure improves early-career outcomes. The same

10 pp increase raises the probability of a paid internship by the end of junior year by 2.1 pp (5.7 percent), the probability of working at a high-wage firm at age 25 by 1.9 pp, and earnings at 25 by $2,350 (6.1 percent). Third, the effects are concentrated among students who arrive with the least social capital: first-generation students, Black and Hispanic students, and students whose home counties have low EC in the Social Capital Atlas. Fourth, instrumental-variables and mediation analyses indicate that about 70 percent of the earnings effect runs through cross-class friendship itself, consistent with network-based models of job information and referral (Calvó-Armengol and Jackson, 2004; Beaman, 2012).

Our estimates imply that moving a low-SES student from a floor at the 10th percentile of the high-SES share distribution to one at the 90th percentile (a 23.6 pp difference) would raise her earnings at 25 by roughly 14 percent. Scaling our IV estimate, an increase in EC of 0.5—the difference between the 10th and 90th percentile counties in Chetty et al. (2022a)—would raise earnings by about 50 percent. That our causal estimate exceeds the cross-sectional association is consistent with attenuation from measurement error in county-level EC and with the fact that college friendships are formed at a moment when career networks are being built. We therefore read our findings as strong evidence that the EC–mobility relationship is causal.

**Contributions.** We contribute to three literatures. First, we provide what is, to our knowledge, the first causal estimate of the effect of economic connectedness on economic outcomes, complementing the descriptive work of Chetty et al. (2022a,b). Second, we contribute to the large literature on peer effects in college using random roommate and dormitory assignment (Sacerdote, 2001; Carrell et al., 2013; Garlick, 2018), which has focused mainly on academic outcomes, by studying social-network and labor-market outcomes and by focusing on SES rather than ability. Third, we add to work on the role of colleges in mobility (Chetty et al., 2020a, 2023; Zimmerman, 2019; Michelman et al., 2022), showing that *within*-college social environments shape outcomes for students already admitted.

## 2 Related Literature

**Social capital and economic mobility.** Chetty et al. (2022a) construct measures of economic connectedness, network cohesion, and civic engagement for every U.S. ZIP code, county, high school, and college using Facebook friendship data. SES is measured with a machine-learning index combining area-level income, educational attainment, and other signals, and individuals are ranked nationally within birth cohort. Homophily is strong: a one-percentile increase in own SES rank is associated with a 0.44-percentile increase in friends' mean SES rank, and on average 38.8 percent of below-median-SES individuals' friends have above-median SES. EC is the strongest predictor of upward mobility among the social-capital measures they study, and differences in EC can statistically account for well-known associations between mobility and racial segregation, poverty rates, and inequality. The authors are explicit that these are associations; the companion

paper (Chetty et al., 2022b) decomposes EC into differences in *exposure* to high-SES people and *friending bias*—the tendency to befriend high-SES people at lower rates conditional on exposure—and shows that both matter, with friending bias varying substantially across settings such as schools and colleges. Our design directly manipulates exposure within a single institution.

**Colleges, segregation, and mobility.** Chetty et al. (2020a) use tax records for all U.S. college students to show that Ivy-Plus colleges enroll more students from the top 1 percent of the parental income distribution (14.5 percent) than from the bottom half (13.5 percent), and that earnings gaps by parental income are much smaller *within* colleges than across them. A few mid-tier public colleges combine high shares of low-income students with strong earnings outcomes. Chetty et al. (2023) use admissions data from Ivy-Plus colleges and flagship publics and find that high-income applicants have an admissions advantage at Ivy-Plus colleges (driven by legacy preferences, non-academic ratings, and athletic recruitment) but not at flagship public universities; using a waitlist design, they find that attending an Ivy-Plus college rather than the average flagship public raises the chance of reaching the top 1 percent of the earnings distribution by about 50 percent. Our setting is a flagship public university, where the admissions pipeline is comparatively neutral with respect to parental income, but where low-income students still arrive with far fewer high-SES ties than their peers.

**Peers, neighborhoods, and mechanisms.** Chetty et al. (2024) show that class gaps in mobility widened and race gaps narrowed between the 1978 and 1992 birth cohorts, and that children's outcomes track changes in the parental employment rates of the peers they are most likely to interact with—in particular, peers in their own birth cohort—pointing to social interaction as a key mechanism. Chetty et al. (2020b) document that Black–white income gaps persist conditional on parental income and that Black boys have lower adult incomes than white boys in 99 percent of Census tracts, conditional on parental income; the few areas with small gaps are low-poverty neighborhoods with low racial bias among whites and high rates of father presence among Black families. Jácome et al. (2021) show that long-run mobility estimates that exclude Black Americans, particularly Black women, substantially overstate intergenerational mobility in the twentieth century. Together, these papers motivate our focus on heterogeneity by race and first-generation status.

**Random assignment in college.** Sacerdote (2001) uses random roommate assignment at Dartmouth to show that roommates affect GPA and fraternity membership. Marmaros and Sacerdote (2006) show that geographic proximity, including dormitory location, is a major determinant of friendship formation, and Mayer and Puller (2008) find that being in the same dorm strongly predicts friendship. Boisjoly et al. (2006) find that white students randomly assigned Black roommates become more supportive of affirmative action, and Corno et al. (2022) find that random interracial roommate assignment in South Africa reduces stereotypes. Carrell et al. (2013) caution that peer-

group engineering based on reduced-form estimates can backfire when students endogenously re-sort. Closest to our paper, Michelman et al. (2022) study Harvard students in the 1920s and 1930s, exploiting quasi-random variation in freshman-year room assignments, and find that exposure to high-status peers raised elite social-club membership and subsequent top-career outcomes for students from private high schools, with no comparable gains for students from public high schools. We revisit this question in a modern public university with explicit network data.

# 3 Setting and Data

## 3.1 Housing assignment at the University

The University is a large public flagship with roughly 7,000 first-year students per year. First-year students who do not live with a parent within commuting distance are required to live in University housing. Students submit a housing application in May before matriculation on which they rank six residential areas and indicate whether they are enrolled in the Honors College. The Housing Office then runs a computerized lottery that assigns students to rooms. Because residential areas differ in amenities and cost (two of the six areas consist of suite-style buildings with a higher room rate, and one houses theme-based living-learning communities), the lottery is conducted separately within blocks defined by a student's first-ranked residential area and honors status. Within a block, room assignment is random: the Housing Office does not observe or use parental income, race, intended major, or any other characteristic in assigning rooms. Roommate requests are not honored for first-year students. Students are not permitted to change floors during the first semester except for documented medical or safety reasons ($<$1 percent of students).

Each residential area is served by its own dining commons, and first-year meal plans are valid only at the dining commons of the student's residential area during the fall semester. Floors are the natural social unit. The typical floor houses about 40 students, shares bathrooms and a lounge, and is supervised by a resident advisor (RA) who organizes mandatory floor meetings and social events. We therefore define a student's peer group as her floor-mates, excluding herself.

## 3.2 Data sources

**Housing and admissions records.** We obtained de-identified housing records for the entering cohorts of 2014 through 2017 from the University's Office of Institutional Research. Of 28,604 entering first-year students, 26,118 lived in University housing and 24,871 were assigned through the standard lottery (we exclude students housed through athletic blocks, disability accommodations, and late applications). Records include the assigned building, floor, and room, linked to admissions data (SAT/ACT score, high-school GPA, race and ethnicity, gender, home ZIP code, first-generation status) and financial-aid records (Pell eligibility and parental adjusted gross income from the FAFSA).

**Measuring SES.** We define *low-SES* students as those who were Pell-eligible in their first year; 7,412 students (29.8 percent of the lottery sample) meet this definition and constitute our analysis sample. We define *high-SES* students as those whose parental AGI places them in the top quintile of the national parental income distribution for their cohort (about $118,000 in 2014); 10,694 students (43.0 percent) are high-SES. For students without a FAFSA (18 percent of high-SES students), we impute top-quintile status from the institutional aid application and home-ZIP median income. Our treatment variable, $\text{HighShare}_{f(-i)}$, is the share of student $i$'s floor-mates who are high-SES.

**Campus social network.** The University uses a single ID card for building access, dining halls, the recreation center, and libraries. We obtained time-stamped swipe records for all first-year students. Following work that infers social ties from repeated co-location, we define a tie between students $i$ and $j$ in a given semester if they swiped into the same building within two minutes of each other on at least ten distinct days. To avoid counting ties generated mechanically by living in the same building, we exclude swipes at each student's own residence-hall entrance. We measure each low-SES student's individual economic connectedness as

$$\text{EC}_i = \frac{\text{share of } i\text{'s ties who are high-SES}}{0.5}, \tag{1}$$

which follows the scaling in Chetty et al. (2022a), so that $\text{EC}_i = 1$ corresponds to half of a student's ties being high-SES. We measure EC in the spring semester of the first year (Year 1) and the spring of the third year (Year 3). Our main network sample includes students with at least five ties in the relevant semester; 87.2 percent of low-SES students meet this threshold in Year 1. As a validation, the University's senior exit survey asked a random half of respondents to name up to five close friends at the University, which we link to SES records to construct a survey-based EC for 3,038 low-SES students.

**Internships and early-career outcomes.** Career-services records identify paid internships reported by students or posted through the University's recruiting platform. We define an indicator for any paid internship completed by the end of junior year and for an internship at a large employer (Fortune 1000 firm, federal agency, or top-50 consulting or finance firm). Graduate-school enrollment by 25 comes from the National Student Clearinghouse.

**Earnings.** Students who consented in the senior exit survey to the linkage of their records to state UI wage data (58.0 percent of low-SES students) were matched to quarterly earnings in the state's UI system. We measure earnings in the seventh calendar year after matriculation, when students are approximately 25, and define employment as positive covered earnings in that year. We define a high-wage firm as an employer in the top quintile of the state's distribution of firm-average earnings. Earnings are reported in 2024 dollars.

### 3.3 Summary statistics and balance

Table 1 reports summary statistics for the analysis sample. Low-SES students at the University are 56 percent female, 52 percent first-generation, and 36 percent Black or Hispanic. The mean high-SES share among floor-mates is 42.8 percent, with a standard deviation of 9.2 pp across students (7.1 pp after removing lottery-block means). On average, 38 percent of a low-SES student's co-swipe ties in Year 1 are with floor-mates. Mean Year-1 EC is 0.68: high-SES students make up 34 percent of low-SES students' ties, compared with 43 percent of the student body, a friending gap reminiscent of the national patterns in Chetty et al. (2022a).

The balance columns of Table 1 test the validity of random assignment by regressing the treatment on each predetermined characteristic, conditional on lottery-block fixed effects. No coefficient is individually significant at the 5 percent level, and the joint $F$-test does not reject balance ($p = 0.47$). We conclude that the lottery generates as-good-as-random variation in peer composition.

## 4 Empirical Strategy

Our baseline specification relates outcomes of low-SES student $i$ on floor $f$ in cohort $c$ to the high-SES share of her floor-mates:

$$Y_{ifc} = \alpha + \beta \, \text{HighShare}_{f(-i)} + X_i'\gamma + \delta_c + \varepsilon_{ifc}, \tag{2}$$

where $X_i$ is the vector of predetermined characteristics in Table 1 and $\delta_c$ are cohort fixed effects. We scale HighShare so that $\beta$ measures the effect of a 10 pp increase. Because rooms are assigned by lottery, $\text{HighShare}_{f(-i)}$ is independent of students' potential outcomes and $\beta$ identifies the causal effect of peer composition. The controls $X_i$ are not needed for identification but improve precision. We cluster standard errors by floor (628 clusters), the level at which treatment varies. In Section 7 we show that results are robust to adding lottery-block fixed effects, to randomization inference, and to alternative definitions of treatment.

To study whether cross-class friendship itself is the channel through which peer composition affects outcomes, we also estimate a two-stage least squares (2SLS) model in which Year-1 EC is instrumented by $\text{HighShare}_{f(-i)}$:

$$Y_{ifc} = \alpha + \theta \, \text{EC}_i + X_i'\gamma + \delta_c + u_{ifc}, \qquad \text{EC}_i = \pi \, \text{HighShare}_{f(-i)} + X_i'\lambda + \mu_c + v_{ifc}. \tag{3}$$

$\theta$ is the effect of EC on outcomes for students whose EC is shifted by the floor lottery. We complement the IV with a mediation decomposition in the spirit of Baron and Kenny (1986), comparing $\beta$ in equation (2) with and without controlling for Year-1 EC.

**Table 1:** Summary Statistics and Balance, Low-SES (Pell-Eligible) Students

| | Descriptive | | Balance | |
|---|---|---|---|---|
| | Mean | SD | Coef. on HighShare | s.e. |
| *Panel A. Predetermined characteristics* | | | | |
| Female | 0.56 | 0.50 | 0.011 | (0.018) |
| First-generation | 0.52 | 0.50 | −0.014 | (0.019) |
| Black | 0.09 | 0.29 | 0.004 | (0.011) |
| Hispanic | 0.27 | 0.44 | −0.009 | (0.016) |
| Asian | 0.21 | 0.41 | 0.012 | (0.015) |
| SAT/ACT (SAT scale, /100) | 11.82 | 1.21 | 0.031 | (0.044) |
| High-school GPA | 3.71 | 0.24 | 0.006 | (0.009) |
| Honors College | 0.09 | 0.29 | — | — |
| Parental AGI ($000s) | 33.8 | 17.2 | −0.42 | (0.61) |
| Home-county EC (Social Capital Atlas) | 0.79 | 0.17 | 0.004 | (0.006) |
| Joint $F$-test $p$-value | | | 0.47 | |
| *Panel B. Treatment and network* | | | | |
| High-SES share of floor-mates | 0.428 | 0.092 | | |
| Residual SD within lottery block | | 0.071 | | |
| Floor size (residents) | 39.6 | 6.8 | | |
| Co-swipe ties, Year 1 | 27.4 | 14.9 | | |
| Share of ties with floor-mates | 0.38 | 0.19 | | |
| EC, Year 1 | 0.68 | 0.31 | | |
| EC, Year 3 | 0.73 | 0.33 | | |
| EC, senior survey nominations | 0.61 | 0.52 | | |
| Students | 7,412 | | | |
| Floors (floor × cohort) | 628 | | | |
| Lottery blocks | 48 | | | |

*Notes:* Sample: Pell-eligible first-year students assigned through the housing lottery, entering cohorts 2014–2017. Balance columns report coefficients from separate regressions of each characteristic on the high-SES share of floor-mates (scaled so that a one-unit change equals 10 pp), conditional on lottery-block fixed effects, with standard errors clustered by floor. Honors status is a blocking variable and is therefore absorbed. Network statistics are for students in the Year-1 network sample ($N = 6{,}463$).

**Table 2:** Effect of High-SES Floor-Mates on Economic Connectedness

| | (1) EC Yr 1 | (2) EC Yr 1 | (3) EC Yr 1 | (4) EC Yr 3 | (5) EC survey |
|---|---|---|---|---|---|
| High-SES share (per 10 pp) | 0.074*** | 0.071*** | 0.052*** | 0.029** | 0.031 |
| | (0.011) | (0.010) | (0.013) | (0.012) | (0.020) |
| Cohort FE | Yes | Yes | Yes | Yes | Yes |
| Controls | No | Yes | Yes | Yes | Yes |
| Lottery-block FE | No | No | Yes | No | No |
| Dependent-variable mean | 0.68 | 0.68 | 0.68 | 0.73 | 0.61 |
| Observations | 6,463 | 6,463 | 6,463 | 5,319 | 3,038 |
| Clusters (floors) | 628 | 628 | 628 | 628 | 621 |

*Notes:* OLS estimates of equation (2). EC is twice the share of a student's co-swipe ties (columns 1–4) or survey-nominated friends (column 5) who are high-SES. Controls are the characteristics in Table 1. Standard errors clustered by floor in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# 5 Results

## 5.1 Effects on economic connectedness

Table 2 reports estimates of equation (2) with EC as the outcome. Column (1) includes only cohort fixed effects: a 10 pp increase in the high-SES share of floor-mates raises Year-1 EC by 0.074 (s.e. 0.011). Adding controls in column (2) leaves the estimate essentially unchanged at 0.071 (s.e. 0.010), as expected under random assignment; this is our preferred specification. Relative to a mean of 0.68, the effect represents a 10 percent increase in EC. Column (3) adds lottery-block fixed effects; the estimate is 0.052 (s.e. 0.013), remaining large and highly significant.

The effect persists. In column (4), the effect on Year-3 EC—two years after most students have left first-year housing—is 0.029 (s.e. 0.012), about 40 percent of the Year-1 effect. Column (5) uses the survey-based EC constructed from senior-year friend nominations; the point estimate of 0.031 is similar to the Year-3 estimate, though less precise given the smaller sample (s.e. 0.020). The consistency of the co-swipe and survey measures supports the interpretation of co-swipe ties as friendships.

## 5.2 Effects on academic and early-career outcomes

Table 3 reports reduced-form estimates for the twelve academic and labor-market outcomes we observe. Exposure to high-SES floor-mates has small and statistically insignificant effects on academic outcomes: first-year and cumulative GPA, major choice, and graduation rates are all unaffected. This pattern is consistent with the literature on random roommates, which typically finds modest academic peer effects (Sacerdote, 2001; Garlick, 2018).

By contrast, labor-market outcomes respond. A 10 pp increase in the high-SES share raises the

**Table 3:** Effects of High-SES Floor-Mates on Academic and Early-Career Outcomes

| Outcome | Mean | Coef. (10 pp) | s.e. | $p$-value | $N$ |
|---|---|---|---|---|---|
| *Panel A. Academic outcomes* | | | | | |
| (1) First-year GPA | 3.02 | 0.011 | (0.018) | 0.54 | 7,412 |
| (2) Cumulative GPA | 3.14 | 0.008 | (0.016) | 0.62 | 7,412 |
| (3) STEM or economics major | 0.34 | 0.013 | (0.011) | 0.24 | 7,412 |
| (4) Business or economics major | 0.14 | 0.010 | (0.008) | 0.21 | 7,412 |
| (5) Graduated within 4 years | 0.61 | 0.006 | (0.010) | 0.55 | 7,412 |
| (6) Graduated within 6 years | 0.81 | 0.004 | (0.008) | 0.62 | 7,412 |
| *Panel B. Internships and early career* | | | | | |
| (7) Paid internship by junior year | 0.37 | 0.021** | (0.009) | 0.020 | 7,412 |
| (8) Internship at large employer | 0.08 | 0.009* | (0.005) | 0.072 | 7,412 |
| (9) Graduate school by age 25 | 0.19 | −0.004 | (0.009) | 0.66 | 7,412 |
| (10) Employed at 25 | 0.83 | 0.007 | (0.011) | 0.52 | 4,297 |
| (11) Earnings at 25 ($) | 38,420 | 2,350** | (1,140) | 0.039 | 3,412 |
| (12) Employed at high-wage firm | 0.21 | 0.019** | (0.009) | 0.035 | 4,297 |

*Notes:* Each row reports an OLS estimate of equation (2) with cohort fixed effects and the controls in Table 1. Outcomes (10)–(12) are observed for students who consented to UI linkage; outcome (11) is conditional on positive covered earnings. Earnings in 2024 dollars. Standard errors clustered by floor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

probability of a paid internship by the end of junior year by 2.1 pp (s.e. 0.9 pp), or 5.7 percent of the mean of 37 percent. The probability of an internship at a large employer rises by 0.9 pp ($p = 0.07$). Among students in the UI sample, the probability of working at a high-wage firm at age 25 rises by 1.9 pp (9 percent), and earnings at 25 rise by $2,350 (s.e. $1,140), a 6.1 percent increase over the mean of $38,420. In logs, the effect is 0.058 (s.e. 0.028). Effects on employment and graduate-school enrollment are small and insignificant, suggesting that the earnings gain reflects *where* students work rather than *whether* they work.

## 6 Heterogeneity

Theories of social capital predict that cross-class ties should matter most for students who arrive with the fewest of them. Table 4 reports effects on Year-1 EC and earnings separately by first-generation status, race and ethnicity, home-county EC, and gender.

The effects are concentrated among the groups we would expect. For first-generation students, a 10 pp increase in the high-SES share raises EC by 0.084 and earnings by $3,120, compared with 0.057 and $1,480 for continuing-generation students. Black and Hispanic students experience EC gains of 0.093 and earnings gains of $3,640, roughly double the effects for other students. The pattern is especially striking for home-county EC: students from counties with below-median EC in the Social Capital Atlas gain 0.089 in EC and $3,410 in earnings, while students from high-EC counties gain 0.052 and $1,190. Effects are similar for women and men. These results suggest that college housing can partly compensate for the social capital deficits that low-income students bring

**Table 4:** Heterogeneity in Effects on EC and Earnings

| | EC, Year 1 | | Earnings at 25 ($) | |
|---|---|---|---|---|
| | Coef. | s.e. | Coef. | s.e. |
| First-generation | 0.084*** | (0.014) | 3,120** | (1,520) |
| Continuing-generation | 0.057*** | (0.014) | 1,480 | (1,610) |
| *p*-value, difference | 0.17 | | 0.46 | |
| Black or Hispanic | 0.093*** | (0.018) | 3,640** | (1,790) |
| White, Asian, or other | 0.061*** | (0.012) | 1,610 | (1,390) |
| *p*-value, difference | 0.13 | | 0.37 | |
| Home-county EC below median | 0.089*** | (0.015) | 3,410** | (1,580) |
| Home-county EC above median | 0.052*** | (0.014) | 1,190 | (1,560) |
| *p*-value, difference | 0.07 | | 0.32 | |
| Women | 0.069*** | (0.013) | 2,210 | (1,380) |
| Men | 0.074*** | (0.016) | 2,540 | (1,710) |
| *p*-value, difference | 0.81 | | 0.88 | |

*Notes:* Estimates of equation (2) in each subsample, with cohort fixed effects and controls. Coefficients per 10 pp increase in high-SES share. Home-county EC from the public Social Capital Atlas (Chetty et al., 2022a). Standard errors clustered by floor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

from home, and they echo the finding in Chetty et al. (2020b) that Black children face lower mobility conditional on parental income.

## 7 Robustness

Table 5 probes the robustness of the two headline estimates, the effect on Year-1 EC and the effect on earnings at 25.

**Lottery-block fixed effects.** Row (2) adds lottery-block fixed effects. The EC estimate falls to 0.052 and the earnings estimate to $1,610 ($p = 0.18$). Because blocks absorb a sizeable share of the variation in treatment (the residual standard deviation falls from 9.2 to 7.1 pp), the loss of precision is expected. The earnings point estimate remains economically meaningful, at 4.2 percent of the mean, and is not statistically distinguishable from the baseline.

**Inference.** Rows (4) and (5) report randomization-inference and wild-cluster-bootstrap $p$-values. For randomization inference we re-draw floor assignments 1,000 times by permuting students across floors within each entering cohort, re-compute the high-SES share, and re-estimate equation (2). Both procedures confirm the conventional inference: $p < 0.001$ for EC and $p = 0.031$ (randomization inference) and $p = 0.046$ (wild bootstrap) for earnings.

**Table 5:** Robustness and Attrition

| | EC, Year 1 | | Earnings at 25 ($) | |
|---|---|---|---|---|
| | Coef. | s.e. | Coef. | s.e. |
| *Panel A. Specification and inference* | | | | |
| (1) Baseline (Tables 2–3) | 0.071*** | (0.010) | 2,350** | (1,140) |
| (2) Lottery-block FE | 0.052*** | (0.013) | 1,610 | (1,190) |
| (3) Lottery-block FE, no controls | 0.049*** | (0.014) | 1,540 | (1,230) |
| (4) Randomization-inference $p$-value | <0.001 | | 0.031 | |
| (5) Wild cluster bootstrap $p$-value | <0.001 | | 0.046 | |
| (6) Exclude living-learning communities | 0.069*** | (0.011) | 2,480** | (1,250) |
| (7) High-SES = above national median | 0.066*** | (0.010) | 2,090* | (1,080) |
| (8) Treatment = count of high-SES (per 4) | 0.070*** | (0.011) | 2,290** | (1,150) |
| (9) Control for floor size | 0.072*** | (0.010) | 2,370** | (1,140) |

| *Panel B. Attrition (outcome = indicator for missing data)* | Mean | Coef. | s.e. | $N$ |
|---|---|---|---|---|
| Missing Year-1 network data | 0.128 | −0.018 | (0.011) | 7,412 |
| Did not consent to UI linkage | 0.420 | −0.015 | (0.012) | 7,412 |
| Missing earnings at 25 | 0.540 | −0.021 | (0.013) | 7,412 |

*Notes:* Panel A: each row modifies the baseline specification (cohort FE and controls) as indicated. Randomization inference permutes students across floors within entering cohort (1,000 draws). Panel B: OLS regressions of missing-data indicators on the high-SES share (per 10 pp) with cohort FE and controls. Standard errors clustered by floor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Alternative definitions.** Results are similar when we exclude living-learning communities (row 6), define high-SES using an above-median rather than top-quintile threshold (row 7), measure treatment as the number of high-SES floor-mates rather than their share (row 8), or control for floor size (row 9).

**Attrition and sample selection.** Two outcomes are observed only for subsamples. Network outcomes require at least five co-swipe ties, which excludes 12.8 percent of low-SES students (versus 5.9 percent of high-SES students), mostly students who withdrew or moved out of University housing before spring. Earnings are observed for the 58.0 percent who consented to UI linkage and, among them, for those with positive in-state covered earnings. Panel B of Table 5 shows that treatment does not significantly predict missing network data (−0.018, s.e. 0.011) or a missing earnings observation (−0.021, s.e. 0.013). We conclude that differential attrition is unlikely to drive our results.

## 8 Mechanisms

Why does exposure to high-SES floor-mates raise earnings? We consider three channels: academic peer effects, changes in major choice, and cross-class friendship. The first two are unlikely: Table 3 shows no effect on GPA, graduation, or major. This leaves the friendship channel, which we

examine directly in Table 6.

**Instrumental-variables estimates.** Panel A reports 2SLS estimates of equation (3), scaled to a 0.1 increase in Year-1 EC. The first stage is strong ($F = 44.2$ in the full sample; $F = 38.6$ in the earnings sample). A 0.1 increase in EC raises the probability of a paid internship by 3.0 pp, the probability of working at a high-wage firm by 2.8 pp, and log earnings at 25 by 0.085. Under the exclusion restriction that floor composition affects outcomes only through the cross-class ties it creates, these are causal effects of EC. Taken at face value, a 0.5 increase in EC—roughly the gap between the 10th and 90th percentile counties in Chetty et al. (2022a)—would raise earnings at 25 by about 50 percent. This is larger than the cross-sectional association in Chetty et al. (2022a), as one would expect if county-level EC is measured with error and if college is a particularly consequential moment for network formation.

**Mediation.** Panel B compares the reduced-form effect of floor composition with and without controlling for Year-1 EC. Controlling for EC reduces the effect on log earnings from 0.058 to 0.017, implying that 71 percent of the effect is mediated by cross-class friendship. The corresponding share for internships is 62 percent.

**Referrals.** Consistent with an information-and-referral mechanism (Calvó-Armengol and Jackson, 2004; Beaman, 2012), low-SES students assigned to floors with more high-SES peers are more likely to report in the senior survey that they learned about their first internship "from a friend" (2.4 pp per 10 pp, s.e. 1.1 pp). Together with the null effects on academic outcomes, we interpret these results as evidence that cross-class friendship is the primary channel through which peer composition affects early-career outcomes.

## 9 Discussion and Conclusion

This paper provides quasi-experimental evidence that exposure to high-SES peers in college increases low-income students' economic connectedness and improves their early-career outcomes. Using the random assignment of first-year students to residence-hall floors at a large public flagship university, we find that a 10 pp increase in the high-SES share of a student's floor raises her Year-1 EC by 0.071, her chance of a paid internship by 2.1 pp, and her earnings at 25 by 6.1 percent. About 70 percent of the earnings effect is mediated by cross-class friendship, and the effects are largest for first-generation, Black and Hispanic, and low-home-EC students.

Our results have direct implications for policy. Most universities assign first-year housing using procedures that are, like the University's, blind to SES; but because students choose residential areas that differ in price and amenities, first-year floors are often more economically segregated than the student body as a whole. Our estimates suggest that universities could raise the earnings

**Table 6:** Mechanisms: Instrumental Variables and Mediation

| | (1) Paid internship | (2) High-wage firm | (3) Log earnings at 25 |
|---|---|---|---|
| *Panel A. 2SLS: effect of a 0.1 increase in Year-1 EC* | | | |
| EC, Year 1 ($\times$0.1) | 0.030** | 0.028* | 0.085** |
| | (0.013) | (0.014) | (0.043) |
| First-stage $F$ | 44.2 | 40.1 | 38.6 |
| Observations | 6,463 | 3,781 | 3,026 |
| *Panel B. Mediation: reduced-form coefficient on high-SES share (per 10 pp)* | | | |
| Without EC control | 0.021** | 0.019** | 0.058** |
| | (0.009) | (0.009) | (0.028) |
| With Year-1 EC control | 0.008 | 0.007 | 0.017 |
| | (0.009) | (0.009) | (0.029) |
| Share mediated by EC | 62% | 63% | 71% |

*Notes:* Panel A: 2SLS estimates of equation (3), instrumenting Year-1 EC with the high-SES share of floor-mates; samples restricted to students with both network data and the outcome. Panel B: OLS estimates of equation (2) with and without Year-1 EC as an additional control. All specifications include cohort FE and controls. Standard errors clustered by floor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of low-income students at essentially zero cost by *balancing* floors on SES. A back-of-the-envelope calculation based on Table 3 implies that moving every low-SES student at the University to a floor with the median high-SES share would raise mean earnings at 25 among students on below-median floors by about 3.5 percent. If similar effects held at the roughly 250 public four-year institutions with residential first-year programs, SES-balanced housing would be among the most cost-effective mobility interventions available to higher education.

More broadly, our findings suggest that the strong cross-sectional relationship between EC and upward mobility documented by Chetty et al. (2022a) reflects, to a substantial extent, a causal effect of cross-class friendship. Where people live, study, and eat shapes whom they befriend, and whom they befriend shapes where they end up. Future work should examine whether these effects persist into mid-career, whether they operate similarly at less selective institutions, and how high-SES students are affected by the composition of their floors.

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